

```

char tsaddress[] = "api.thingspeak.com";
String ThingSpeakAPI = "<ThingSpeak Write Key>";
const int UpdateInterval = 30 * 1000;
long ConnectionTime = 0;
boolean connectedlast = false;
int fcounter = 0;
float value = 100;
WiFiClient client;
void setup()
{
  Serial.begin(9600);
  startEthernet();
}
void loop()
{
  String RecordedAnalogInput0 = String(value++, DEC);
  if (client.available())
  {
    char c = client.read();
  }
  if (!client.connected() && connectedlast)
  {
    client.stop();
  }
  if(!client.connected() && (millis() - ConnectionTime >
UpdateInterval))
  {
    updateThingSpeak("field1="+RecordedAnalogInput0);
  }
  if (fcounter > 3) {startEthernet();}
  connectedlast = client.connected();
}
void updateThingSpeak(String tsData)
{
  if (client.connect(tsaddress, 80))
  {
    client.print("POST /update HTTP/1.1\n");
    client.print("Host: api.thingspeak.com\n");
    client.print("Connection: close\n");
    client.print("X-THINGSPEAKAPIKEY: "+ThingSpeakAPI+"\n");
    client.print("Content-Type: application/x-www-form-
urlencoded\n");
    client.print("Content-Length: ");
    client.print(tsData.length());
    client.print("\n\n");
    client.print(tsData);
    ConnectionTime = millis();
    if (client.connected())
    {
      Serial.println("Connecting to ThingSpeak...");
      Serial.println();
      fcounter = 0;
    }
  }
  else

```

```

{
  fcounter++;
  Serial.println("Connection to ThingSpeak failed
("+String(fcounter, DEC)+")");
  Serial.println();
}
}
else
{
  fcounter++;
  Serial.println("Connection to ThingSpeak Failed
("+String(fcounter, DEC)+")");
  Serial.println();
  ConnectionTime = millis();
}
}
void startEthernet()
{
  fcounter = 0;
  client.stop();
  Serial.println("Connecting");
  WiFi.begin(mynetwork, mypassword);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500); }
}

```

Transferring data to Carriots

Carriots (<https://www.carriots.com>) by Altair is another open IoT platform for logging and processing of sensor data on the cloud based environment. The use cases of Carriots include implementation in smart retail, smart cities, smart energy, smart oil, smart agriculture, smart buildings, smart banking and others.

The Carriots platform can be accessed after free registration at *carriots.com*. After logging into Carriots, a secured control

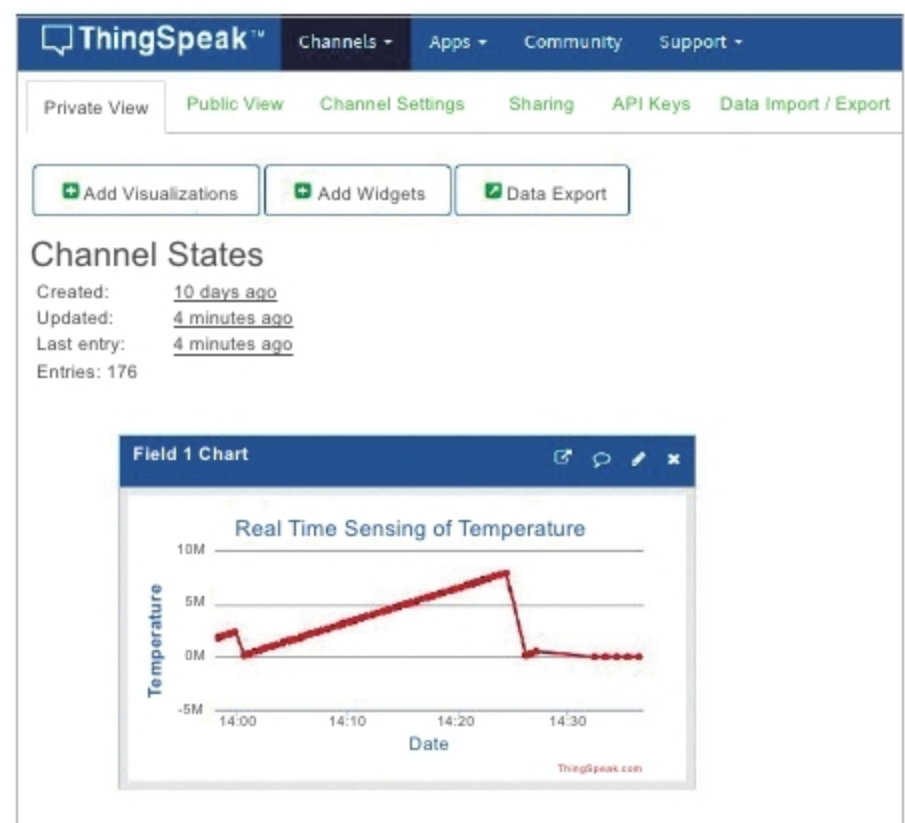


Figure 6: Visualisation of recorded data on ThingSpeak